📖 SCAFFOLDED WORKBOOK · YEAR 9-10

ADVANCED WATER SAFETY

Health & Physical Education · Year 9–10 · Victorian Curriculum F–10 Version 2.0

This workbook has **word banks**, **sentence starters**, and **structured frameworks** to help you. The questions are the same as the standard workbook — the scaffolds just make it easier to get started.

STUDENT NAME	CLASS	TEACHER

📌 **How the scaffolds work**

- **Word banks** — give you the key terms to choose from so you don't have to think of them yourself
- **Sentence starters** — begin the sentence for you; just finish it in your own words
- **Argument frameworks** — break an argument into parts (claim, evidence, explanation) so you can build it step by step
- **Calculation guides** — show you the formula and what to substitute in

1 Advanced Data Analysis — Victorian & Global Drowning Trends

📋 **KEY DATA — KEEP THIS HANDY**

📊 **54 people** drowned fatally in Victoria in 2023–24 — the highest toll in over a decade.

🧮 Victoria's population ≈ **6.9 million**. Rate formula: deaths ÷ (population ÷ 100,000). So: 54 ÷ 69 = **0.78 per 100,000**.

AU Australia's national drowning rate ≈ **1.2 per 100,000** (RLSSA). Victoria is actually BELOW the national average.

🌏 CALD communities = approx. **40%** of Victorian drowning deaths, but only approx. **28%** of the population.

🧑 **85%** of Victorian drowning deaths (46 of 54) were male.

🚫 **All 54 deaths** occurred at unpatrolled locations. Zero deaths at patrolled beaches or supervised pools.

Activity 1.1 — Calculations & Interpretation

~15 min

Question 1.1a — Calculation

2 marks

Calculate the percentage of Victorian drowning deaths that were male. Show your working using the formula below.

CALCULATION FORMULA

Percentage = (number of males ÷ total deaths) × 100

= (_______ ÷ 54) × 100 = _______%

My answer: _______% of Victorian drowning deaths were male.

Now compare to the global figure of ~80%. Is Victoria higher or lower?

Victoria's rate is higher / lower (circle one) than the global figure of 80%.

This suggests that ___________________________________

Question 1.1b — Rate Comparison

3 marks

Calculate: if Victoria's drowning rate matched the national average of 1.2 per 100,000, how many deaths would be expected? Show working.

CALCULATION GUIDE

Expected deaths = rate × (population ÷ 100,000)

= 1.2 × _______ (units of 100,000 in 6.9M) = _______ deaths

At the national rate, Victoria would expect approximately _______ deaths per year.

This is more / fewer (circle) than the actual figure of 54 deaths.

What this tells us about Victoria's performance: ____________________

Question 1.1c — CALD Analysis

3 marks

CALD communities = 40% of drowning deaths but 28% of population. Are they over-represented or under-represented? Explain TWO structural reasons.

WORD BANK

over-represented | structural | swim lessons | language barriers | fishing risk | 40% > 28%

CALD Victorians are _____________ in drowning deaths because 40% of deaths > 28% of population.

Reason 1 — Use this argument framework:

REASON: ______________________________

BECAUSE: ______________________________

THIS MEANS: ______________________________

Reason 2 — Use this argument framework:

REASON: ______________________________

BECAUSE: ______________________________

THIS MEANS: ______________________________

2 Physiology of Drowning — What Happens in the Body

📋 KEY FACTS — PHYSIOLOGY

- 🧠 **4–6 minutes** — after this long without oxygenated blood, brain cells start dying irreversibly. This is called **cerebral hypoxia**.
- 💓 CPR generates about **25–30%** of normal cardiac output — enough to keep the brain alive until the heart can restart.
- 🥶 **1-10-1 Cold Water Rule:** ~1 minute to control breathing · ~10 minutes of useful swimming · up to 1 hour before hypothermia.
- 💧 **Secondary drowning** — fluid in the lungs can develop HOURS after a near-drowning. Always get a medical check after any submersion, even if the person feels fine.

Activity 2.1 — Physiology Research Tasks

~15 min

Question 2.1a — Physiological Sequence — 3 marks

Put the following events in the correct order from 1 (first) to 5 (last):

EVENTS TO ORDER

cardiac arrest · submersion · laryngospasm · hypoxia · cerebral hypoxia

Order	Event	What it means
1 (first)	______________________	Person goes underwater
2	______________________	Airway closes involuntarily
3	______________________	Oxygen levels in blood drop

4	____________________	Heart stops pumping effectively
5 (last)	____________________	Brain cells start dying — time-critical!

Question 2.1b — Why Start CPR Quickly? 3 marks

Use the sentence starters below to explain the 4-minute window in plain language.

When the heart stops, the brain stops getting ________________________ blood.

After about _______ minutes without oxygen, brain cells start ________________________ .

CPR generates about ___% of normal cardiac output, which is ________________________ to keep the brain alive.

This is why starting CPR within 4 minutes is so important:

____

Question 2.1c — Secondary Drowning 2 marks

Explain secondary drowning and why it matters. Use the word bank.

WORD BANK

hours later | fluid in the lungs | coughing | medical review | near-drowning | rare but serious

Secondary drowning is when ________________________ develops ________________________ a near-drowning incident.

It is clinically important because ________________________________

After any submersion incident, even if the person seems fine, they should ________________________ .

Source URL: ____

3 Social Determinants of Drowning Risk

Activity 3.1 — Social Determinants Analysis

Pairs · ~18 min

Social determinants are **non-medical factors** that influence health outcomes — things like where you grew up, your income, and cultural norms.

Question 3.1a

4 marks

Choose TWO social determinants from the list below. For each, complete the argument framework.

SOCIAL DETERMINANTS — CHOOSE 2

Gender norms (masculinity) | CALD background | Socioeconomic status | Rural / regional location | Alcohol use

Social Determinant 1: ___________________

HOW IT INCREASES RISK: ______________________________

EVIDENCE / EXAMPLE: ______________________________

PREVENTION STRATEGY: ______________________________

Social Determinant 2: ___________________

HOW IT INCREASES RISK: ______________________________

EVIDENCE / EXAMPLE: ______________________________

PREVENTION STRATEGY: ______________________________

Question 3.1b — Individual vs Structural

2 marks

What is the difference between an **individual** explanation and a **structural** explanation for a health problem? Give one example of each using the CALD drowning data.

DEFINITION BANK

personal choice | system-level | lack of swim access | didn't learn to swim

An individual explanation blames _________________________ choices. Example: "They _________________________ ."

A structural explanation focuses on _________________________ factors. Example: "In many countries of origin, _________________________ ."

Structural explanations are more useful for prevention because _________________________________

Question 3.1c — School Actions 3 marks

Propose THREE specific, practical actions a Victorian school could take to reduce CALD drowning disproportionality.

Tip: be SPECIFIC — not just "raise awareness." Think: who, what, where, when, how?

Action 1: The school could _________________________________

Action 2: The school could _________________________________

Action 3: The school could _________________________________

4 Evaluating Drowning Prevention Strategies

Activity 4.1 — Prevention Strategy Evaluation

~15 min

For each strategy, decide if the evidence is Strong, Moderate, or Limited, and explain a limitation.

EVIDENCE STRENGTH — CHOOSE ONE

Strong — backed by solid research | Moderate — some research, some gaps | Limited — little evidence, or limited reach

Strategy	Evidence strength	Key limitation (write in your own words)
4-sided pool fencing		
Mandatory swim lessons for primary school students		
Expanding patrolled beach patrols		
Warning signs at unpatrolled beaches		
Targeted CALD community swim programs		

Question 4.1a — Justify Your Priorities

4 marks

If you had $5 million to spend on drowning prevention, which TWO strategies would you choose? Use the argument framework.

Strategy 1: ____________________

I WOULD CHOOSE THIS BECAUSE: ____________________

THE EVIDENCE SHOWS: ____________________

THE TRADE-OFF IS: ____________________

Strategy 2: ____________________

I WOULD CHOOSE THIS BECAUSE: ____________________

THE EVIDENCE SHOWS: ____________________

THE TRADE-OFF IS: ____________________

5 International Comparison

Activity 5.1 — Research Comparison

~20 min · Devices needed

Visit each website and complete the table. Look for: program name, key safety message, and what makes them different.

WEBSITES TO VISIT

AU royallifesaving.com.au

GB rlss.org.uk/education

US redcross.org/take-a-class/water-safety

Question	AU RLSSA (Australia)	GB RLSS UK	US Red Cross (USA)
Main program name			
Key safety message / slogan			
One thing unique about their approach			

Question 5.1a — Should Victoria Adopt It? 3 marks

Choose ONE element from an international approach that is NOT currently used in Victoria. Should Victoria adopt it? Use the argument framework.

The element I chose: ___________________ From: ___________________ (country/organisation)

WHAT IT IS:

WHY IT COULD HELP IN VICTORIA:

I THINK VICTORIA SHOULD / SHOULD NOT (CIRCLE) ADOPT IT BECAUSE:

SOURCE URL:

6 Diving, Spinal Injuries & High-Risk Teenage Behaviour

📋 **KEY FACTS — KEEP THESE HANDY FOR YOUR ANSWERS**

Word Bank: cervical spine · quadriplegia · spinal cord · sandbank · tidal · sediment · false confidence · masculinity norms · peer pressure · AIHW · RLSSA · feet first · permanent disability · inland waterway

- Unsafe diving = **10%** of all spinal cord injuries and **20%** of quadriplegia cases in Australia (AIHW)
- **95%** of aquatic spinal injury patients are male — **75%+** are under 30 — **66%** suffer permanent disability
- Highest incidence of diving spinal injury: children aged **10–14**
- RLSSA: **"More people drown in inland waterways than any other location in Australia."**
- Core prevention message: **"Feet first, first time, every time."** Never dive into unknown water.
- +34% increase in drowning among 15–20 year olds post-COVID vs pre-COVID (RLSSA)

Sources: Royal Life Saving Australia (royallifesaving.com.au) · AIHW Diving Injury Report (aihw.gov.au) · Royal Perth Hospital spinal injury study

Activity 6.1 — Case Study Analysis

~18 min

CASE STUDY 1 · NSW 2025

Terrigal Wharf

Teenage boy dived from a wharf into shallow water. Helicopter evacuation to hospital. He didn't know how shallow it was under the structure.

CASE STUDY 2 · VICTORIA 2025

Torquay Front Beach

Teenage boy suffered suspected spinal injury diving at a surf beach. Airlifted to hospital. Sandbanks at surf beaches shift with every tide.

CASE STUDY 3 · NSW 2024

Bawley Beach Jetty

Young man seriously injured diving from a jetty. A sandbank had recently formed beneath the jumping spot. He had jumped there safely before.

Question 6.1a — Common Factors

3 marks

Identify two common factors across all three case studies. For each, use the framework below.

Factor 1 — Use this framework:

FACTOR: ____________________

HOW IT CAUSED THE INJURY (NOT JUST THAT IT WAS THERE): ____________________

WHICH CASE STUDY SHOWS THIS MOST CLEARLY:

Factor 2 — Use this framework:

FACTOR:

HOW IT CAUSED THE INJURY:

WHICH CASE STUDY SHOWS THIS:

Question 6.1b — Why 95% of Patients Are Male

4 marks

Use the social determinants framework from Section 3. Don't just say "males take more risks." Explain WHY they take more risks using social factors.

Argument framework:

SOCIAL NORM / EXPECTATION THAT DRIVES RISK-TAKING:

HOW THIS LEADS SPECIFICALLY TO DIVING BEHAVIOUR:

WHY TELLING INDIVIDUALS TO "BE CAREFUL" IS NOT ENOUGH:

WHAT STRUCTURAL CHANGE WOULD ADDRESS THE ROOT CAUSE:

Question 6.1c — Why Neck Injuries Cause Permanent Disability

3 marks

Use the physiology content from Section 2 to explain why a cervical (neck) spine injury from diving often causes permanent quadriplegia.

Sentence starters:

"The cervical spine is responsible for ___________. When a diver hits the bottom head-first, the vertebrae ___________. This is different from other spine levels because ___________. The result is often permanent because ___________."

Activity 6.2 — Myth vs Evidence

~15 min

Use the sentence starters below. Replace the blanks with your own words using facts from the key data section above.

Myth: "If other people are jumping there, it must be safe."

Starter: "This is a myth because depth can change due to ___________ and ___________. The Bawley Beach case shows that a ___________ had formed between visits, making a safe spot suddenly dangerous."

Myth: "I've jumped here before — I know this spot."

Starter: "Prior experience creates false confidence because ___________ changes between visits. The correct approach is to ___________."

Myth: "Good swimmers are safe from diving injuries."

Starter: "Swimming ability is irrelevant to diving injuries because the injury happens ___________ before swimming matters. Australian data shows ___________% of patients were healthy young males."

Myth: "Jumping feet-first is safe — you won't hit your head."

Starter: "Feet-first jumping into shallow water can still cause ___________ and ___________. The key prevention message is ___________."

Activity 6.3 — Prevention Strategy Design

~15 min

My Prevention Strategy

Step 1 — Identify your target group:

TARGET GROUP (BE SPECIFIC — AGE, GENDER, LOCATION):

STATISTIC SHOWING THIS GROUP IS AT HIGHER RISK (CITE SOURCE):

SECOND STATISTIC:

Step 2 — Describe your strategy:

WHAT WILL YOU DO:

WHERE / THROUGH WHAT CHANNEL:

WHO DELIVERS IT:

SMART METRIC — HOW WILL YOU KNOW IF IT WORKED:

Step 3 — Limitation:

THE BIGGEST WEAKNESS OF MY STRATEGY IS:

A MORE COMPLETE RESPONSE WOULD INCLUDE:

7 Assessment Task Planning

Activity 6.1 — Plan Your Assessment Task

Assessment preparation

Choose your option and use the planning scaffold to get started. Remember: all options need at least THREE sources with URLs and at least ONE statistic.

Option A — Policy Brief
600–800 words to Victorian Minister. Two recommendations + evidence + success metrics.

Option B — Campaign
Target ONE high-risk group. Campaign name, message, visual, platform, 100-word justification.

Option C — School Audit ★
Audit your school. Benchmark against Curriculum 2.0. Priority matrix + recommendations.

My Planning Notes

OPTION CHOSEN:

WHO AM I WRITING FOR (AUDIENCE / ADDRESSEE)?

MY KEY ARGUMENT / CENTRAL CLAIM:

STATISTIC I WILL USE:

SOURCE 1 URL:

SOURCE 2 URL:

SOURCE 3 URL:

My draft — write here or attach on a separate page:

✏️ **How Did I Go? Circle: 🙁 Not yet / 🙂 Getting there / 😊 Got it!**

I can calculate Victorian drowning rates and compare them to other benchmarks	🙁 🙂 😊
I can explain the physiological sequence of drowning and why CPR timing matters	🙁 🙂 😊
I can identify social determinants of drowning risk and explain how they work	🙁 🙂 😊
I can evaluate prevention strategies and justify my priorities with evidence	🙁 🙂 😊
I can compare at least two international water safety approaches	🙁 🙂 😊
I have a clear plan for my assessment task with at least 3 sources	🙁 🙂 😊

One thing I still want help with: